5 Generation guarantee of reverse process

In this section, we first consider the reverse process as in (19), called the *exact* reserves process, when there is no inversion error. In Section 5.1, we prove a KL (and TV) guarantee of generating by q_0 any data distribution P in $\mathcal{P}_2^r$, and extend to P with no density by introducing a short-time initial diffusion.

Taking into account the inversion error, we consider the sequence $\tilde{q}_n$ induced by S_n in (32), called the *computed* reverse process, where the inversion error satisfies Assumption 2. In Section 5.2, we prove a closeness bound of $\tilde{q}_0$ to q_0 in $\mathcal{W}_2$, which leads to a $\mathcal{W}_2$-KL mixed generation guarantee of $\tilde{q}_0$ based on the proved guarantee of q_0.

5.1 Convergence guarantee without inversion error

We start by presenting convergence analysis assuming there are no errors in the reverse process; then, we extend to the more practical situation considering inversion errors. All proofs in this section are given in Appendix C.

5.1.1 KL (TV) guarantee of generating P with density

Consider the transport map over the N steps in (19) denoted as

$$T_1^N := T_N \circ \cdots \circ T_1. \tag{38}$$

Since each T_n is invertible (Assumption 1), the overall mapping T_1^N is also invertible. We have

$$p_N = (T_1^N)_\# p_0, \quad q_N = (T_1^N)_\# q_0.$$

The following lemma follows from the data processing inequality of KL, which allows us to obtain KL bound of $p_0 = p$ and q_0 from that of p_N and $q_N = q$.

Lemma 5.1 (Bi-direction data processing inequality)**.** *If $T : \mathbb{R}^d \to \mathbb{R}^d$ is invertible and for two densities p and q on $\mathbb{R}^d$, $T_\# p$ and $T_\# q$ also have densities, then*

$$\mathrm{KL}(p||q) = \mathrm{KL}(T_\# p||T_\# q).$$

The following corollary establishes an $O(\varepsilon^2)$ KL bound in $N \lesssim \log(1/\varepsilon)$ JKO steps, which implies an $O(\varepsilon)$ TV bound by Pinsker's inequality.

Corollary 5.2 (KL guarantee for $P \in \mathcal{P}_2^r$)**.** *Suppose $G(\rho) = \mathrm{KL}(\rho||q)$, the potential function V satisfies Assumption 4 with $\lambda \in (0, 1]$, and $0 < \gamma < 2$. Suppose $P \in \mathcal{P}_2^r$ with density p, let $p_0 = p$, and Assumption 1 holds for some ε and all n. Then, let*

$$N = \left\lceil \frac{8}{\gamma\lambda} (\log \mathcal{W}_2(p_0, q) + \log(\lambda/\varepsilon)) \right\rceil, \tag{39}$$

the generated density q_0 of the reverse process satisfies that

$$\mathrm{KL}(p||q_0) \le \frac{9}{2\gamma} \left(\frac{\varepsilon}{\lambda}\right)^2, \quad \mathrm{TV}(p, q_0) \le \frac{3}{2\sqrt{\gamma}} \frac{\varepsilon}{\lambda}. \tag{40}$$

Remark 5.1 (Extension to f-divergence). As has been discussed in Section 4.1, it is possible to show that $G(\rho) = D_f(\rho||q)$ satisfies Assumption 3 (possibly under additional conditions of f), and then the convergence of the forward process, Theorem 4.3, extends to such f-divergence G. In addition, data processing inequality holds generally for f-divergence, and thus Lemma 5.1 also extends. As a result, Corollary 5.2 can potentially extend to certain f-divergences and show a guarantee of $D_f(p||q_0)$ in N JKO steps.

5.1.2 Guarantee of generating $P \in \mathcal{P}_2$ up to initial short diffusion

For $P \in \mathcal{P}_2$ that may not have a density, we first obtain $\rho_\delta \in \mathcal{P}_2^r$ that is close to P in $\mathcal{W}_2$ by a short-time initial diffusion (specifically, the OU process as introduced in Section 2.3) up to time $\delta > 0$, as shown in Lemma C.1. The short-time initial diffusion was used in [48] and called "early stopping" in [14]. It is also used in practice by flow model [74] as well as score-based diffusion models to bypass the irregularity of data distribution [67]. In principle, one can also use the Brownian motion only (corresponding to convolving P with Gaussian kernel) to obtain ρ_δ. Here we use the OU process to stay in line with the literature.

The introduction of ρ_δ allows us to prove a guarantee of $\mathrm{KL}(\rho_\delta||q_0)$ in the following corollary, which is the same type of result as [14, Theorem 2].

Corollary 5.3 (KL guarantee for $P \in \mathcal{P}_2$ from ρ_δ)**.** *Suppose $P \in \mathcal{P}_2$, and the conditions on G, V, λ and γ are the same as in Corollary 5.2. Then $\forall \varepsilon' > 0$, there exists $\delta > 0$ s.t. $\mathcal{W}_2(P, \rho_\delta) < \varepsilon'$ and, with $p_0 = \rho_\delta$ and Assumption 1 holds for some ε and all n, let N as in* (39)*, the generated density q_0 of the reverse process makes* $\mathrm{KL}(\rho_\delta||q_0)$ *and* $\mathrm{TV}(\rho_\delta, q_0)$ *satisfy the same bounds as in* (40).

The corollary shows that there is a density $\rho_\delta \in \mathcal{P}_2^r$ that is arbitrarily close to P in $\mathcal{W}_2$, such that the output density q_0 of the reverse process can approximate ρ_δ up to the same error as in Corollary 5.2. Note that the corollary holds when the potential function V of q satisfies the general condition Assumption 4. The OU process in the proof (Lemma C.1) is only used in constructing ρ_δ, and there are other means to construct the surrogate initial density ρ_δ.

5.2 Convergence guarantee with inversion error

Recall the set-up from Section 3.3. To prove the $\mathcal{W}_2$ control between $\tilde{q}_0$ and q_0, we first introduce a Lipschitz condition on the inverse of the learned transport map T_n and explain the motivation.

5.2.1 Lipschitz constant of computed transport maps

Previously in Assumption 1, we required that both T_n and its inverse are non-degenerate. Here, we further require that T_n^{-1} is globally Lipschitz on $\mathbb{R}^d$ with a uniform Lipschitz constant.

Assumption 6 (Lipschitz condition on T_n^{-1})**.** *There is $K > 0$ s.t. T_n^{-1} is Lipschitz on $\mathbb{R}^d$ with Lipschitz constant $e^{\gamma K}$ for all $n = N, \dots, 1$.*

The assumed Lipschitz constants are theoretical and motivated by neural ODE models, to be detailed below. Our analysis of $\mathcal{W}_2(\tilde{q}_0, q_0)$ applies to any type of flow network (like invertible ResNet) as long as the needed assumptions on T_n and S_n hold.

We justify the assumptions on T_n, T_n^{-1} and S_n under the framework of neural ODE flow, namely (20)(21), including the Lipschitz constant $e^{\gamma K}$ of T_n^{-1}. Specifically, by the elementary Lemma C.2 proved in Appendix C, we know that if T_{n+1} can be numerically exactly computed as (21) and $\hat{v}(x,t)$ on $\mathbb{R}^d \times [t_n, t_{n+1}]$ satisfies a uniform x-Lipschitz condition with Lipschitz constant K, then both T_{n+1} and its inverse are Lipschitz on $\mathbb{R}^d$ with Lipschitz constant $e^{\gamma K}$. We will assume the same K throughout time for simplicity. In practice, Lipschitz regularization techniques can be applied to the neural network parametrized $\hat{v}(x,t)$, and the global Lipchitz bound of $\hat{v}$ on $\mathbb{R}^d$ can be achieved by "clipping" $\hat{v}$ to vanish outside some bounded domain of x. Meanwhile, note that if $T : \mathbb{R}^d \to \mathbb{R}^d$ is invertible and T^{-1} is globally Lipschitz on $\mathbb{R}^d$, then T is non-degenerate. Thus T_{n+1} and T_{n+1}^{-1} both being non-degenerate are implied by (and weaker than) the global Lipschitzness of T_{n+1} and its inverse. In addition, in a neural-ODE-based flow model, the reverse process is by integrating the neural ODE in reverse time, and thus we can expect similar properties of S_n.

While the computed transport T_n and S_n often differ from the exact numerical integration of the ODE, we still expect the Lipschitz property to retain. For general flow models, which may not be neural ODE, we impose the same theoretical assumptions. At last, the global Lipschitz condition may be theoretically relaxed by combining with truncation arguments of the probability distributions, which is postponed here.

5.2.2 $\mathcal{W}_2$-control of the computed reverse process from the exact one

Proposition 5.4. *Suppose in* (32)*, $q_N = \tilde{q}_N = q \in \mathcal{P}_2^r$, and the computed transport maps T_n and S_n satisfy Assumptions 1, 2 and 6. Then all q_n and $\tilde{q}_n$ are in $\mathcal{P}_2^r$ and*

$$\mathcal{W}_2(\tilde{q}_0, q_0) \le \frac{\varepsilon_{\rm inv}}{\gamma K} e^{\gamma K(N+1)}. \tag{41}$$

A continuous-time counterpart of Proposition 5.4 was derived in [3, Proposition 3]. We include a proof in Appendix C for completeness. The proof uses a coupling argument of the (discrete-time) ODE flow, which as has been pointed out in [16], obtains a growing factor $e^{\gamma KN}$ in the $\mathcal{W}_2$-bound as shown in (41). To overcome this exponential factor, [16] adopted an SDE corrector step. Here, without involving any corrector step, we show that the factor $e^{\gamma KN}$ can be controlled at the order of some negative power of ε thanks to the exponential convergence in the forward process. This is because N can be chosen to be at the order of $\log(1/\varepsilon)$ as in (39), then $e^{\gamma KN}$ can be made $O(\varepsilon^{-\alpha})$ for some $\alpha > 0$. As a result, the $\mathcal{W}_2$-error (41) can be suppressed if $\varepsilon_{\rm inv}$ can be made smaller than a higher power of ε.

More specifically, combined with the analysis of the forward process, we arrive at the following guarantee, proved in Appendix C.

Corollary 5.5 (Mixed bound with inversion error)**.** *Suppose $G(\rho) = \mathrm{KL}(\rho||q)$, the potential function V satisfies Assumption 4 with $\lambda \in (0,1]$, and $0 < \gamma < 2$. Suppose the computed transports maps T_n and S_n satisfy the Assumptions 1, 2, 6 for some ε and $\varepsilon_{\rm inv}$ for all n. Suppose $P \in \mathcal{P}_2^r$ with density p, let $p_0 = p$ and N as in* (39)*, then the generated density $\tilde{q}_0$ of the computed reverse process satisfies that*

$$\mathcal{W}_2(\tilde{q}_0, q_0) \le \frac{e^{2\gamma K}}{\gamma K} (\mathcal{W}_2(p_0, q)\lambda)^{8K/\lambda} \frac{\varepsilon_{\rm inv}}{\varepsilon^{8K/\lambda}}, \tag{42}$$

and q_0 satisfies the KL and TV bounds to p as in (40)*.*

Remark 5.2 ($O(\varepsilon)$ error and need for small $\varepsilon_{\rm inv}$). The corollary implies that if $\varepsilon_{\rm inv}$ can be made small, then the $\mathcal{W}_2$ bound can be made equal to or smaller than ε in order. For example, if $\varepsilon_{\rm inv} = O(\varepsilon^{8K/\lambda+1})$, then we have $\mathcal{W}_2(\tilde{q}_0, q_0) = O(\varepsilon)$. This suggests that if one focuses on getting $\mathrm{KL}(p_N||q)$ small in the forward process, then maintaining an inversion error small is crucial for the generation quality of the flow model in the reverse process.

At last, when P is merely in $\mathcal{P}_2$ and does not have density, then one can start the forward process from $p_0 = \rho_\delta$ same as in Section 5.1.2. Then we have the same $\mathcal{W}_2$-bound between $\tilde{q}_0$ and q_0 as in (42), and q_0 is close to ρ_δ in the sense of Corollary 5.3.

6 Discussion

The work can be extended in several directions. First, it is interesting to see if the assumption on learning in the forward process, Assumption 1, can be derived from further analysis of the neural network learning, e.g., the approximation and optimization error (c.f. the list of sources of errors in